60. MANUFACTURING QUALITY CONTROL MATRIX

1. AIM

To write a Python program using nested loops to evaluate defect percentages across manufacturing production batches and flag individual batches exceeding quality tolerance thresholds.

2. PROBLEM STATEMENT

Manufacturing quality assurance relies on monitoring defective rate percentages across production batches. Write a Python program that accepts product batch details, calculates defective unit percentage, and flags batches requiring quality checks.

3. OBJECTIVES

• Accept number of products, batches per product, and maximum allowed defect limit (in %).
• Iterate using nested loops through products and batches.
• Calculate batch defect percentage: (defective * 100) / made.
• Display quality status ('Quality check needed' vs 'Quality is acceptable').

4. ASSUMPTIONS & REQUIREMENTS

Assumptions: Production count (made) and defective units are non-negative integers.
Requirements: Python 3.x Interpreter, PyCharm environment.

5. ALGORITHM

Step 1: Input products, batches, and defect limit.
Step 2: Loop i from 1 to products.
Step 3: Loop j from 1 to batches: Input made and defective. Compute percent. Display defect %, and check against limit.
Step 4: Stop program.

6. PROGRAM FLOWCHART

7. PYTHON PROGRAM (Manufacturing Quality Control Matrix.py)

```
products = int(input("Enter number of products: "))
batches = int(input("Enter number of batches: "))
limit = float(input("Enter defect percentage limit: "))

for i in range(1, products + 1):
    for j in range(1, batches + 1):
        made = int(input("Enter units produced: "))
        defective = int(input("Enter defective units: "))

        if made > 0:
            percent = defective * 100 / made
        else:
            percent = 0

        print("Product:", i, "Batch:", j)
        print("Defect percentage:", round(percent, 2), "%")

        if percent > limit:
            print("Quality check needed")
        else:
            print("Quality is acceptable")
```

8. EXPLANATION OF THE CODE

• percent = defective * 100 / made: Calculates percentage of defective items.
• round(percent, 2): Rounds output defect rate to two decimal places.

9. TRACING TABLE

Product #	Batch #	Units Made	Defective	Defect %	Limit	Status Flag
Product 1	Batch 1	12	1	8.33%	2.0%	Quality check needed

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Exceeds Limit)	Products: 3, Batches: 3, Limit: 2, Made: 12, Defective: 1	Product: 1 Batch: 1 Defect percentage: 8.33 % Quality check needed	Product: 1 Batch: 1 Defect percentage: 8.33 % Quality check needed

11. EXPECTED OUTPUT

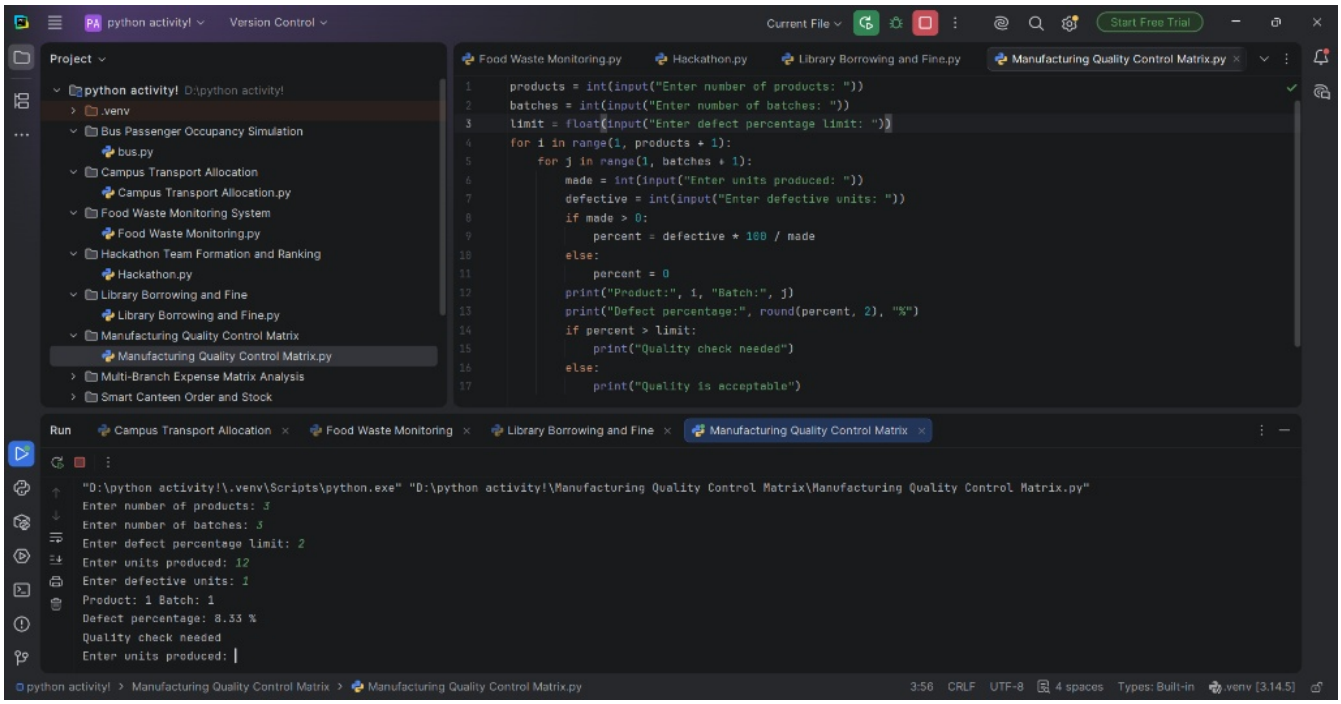
```
Enter number of products: 3
Enter number of batches: 3
Enter defect percentage limit: 2
Enter units produced: 12
Enter defective units: 1
Product: 1 Batch: 1
Defect percentage: 8.33 %
Quality check needed
```

12. ACTUAL OUTPUT

```
Enter number of products: 3
Enter number of batches: 3
Enter defect percentage limit: 2
Enter units produced: 12
Enter defective units: 1
Product: 1 Batch: 1
Defect percentage: 8.33 %
Quality check needed
Enter units produced: |
```

13. PROGRAM EXECUTION SCREENSHOT

Actual Input and Output - PyCharm Run Screenshot



14. RESULT/CONCLUSION

The manufacturing quality control program was executed successfully, calculating defect ratios per batch and flagging batches exceeding tolerance thresholds.